Q1. Create a New Database and Table for Employees

```sql
CREATE DATABASE company_db;

USE company_db;

CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    department VARCHAR(50),
    salary INT,
    hire_date DATE
);
```

Q2. Insert Data into Employees Table

```sql
INSERT INTO employees (employee_id, first_name, last_name, department, salary, hire_date) VALUES
(101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),
(102, 'Riya', 'Kapoor', 'Sales', 75000, '2019-03-22'),
(103, 'Raj', 'Mehta', 'IT', 90000, '2018-07-11'),
(104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),
(105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');
```

Q3. Display All Employee Records Sorted by Salary (Lowest to Highest)

```sql
SELECT *
FROM employees
ORDER BY salary ASC;
```

Q4. Show Employees Sorted by Department (A–Z) and Salary (High to Low)

```sql
SELECT *
FROM employees
ORDER BY department ASC, salary DESC;
```

Q5. List All Employees in IT Department (Newest First)

```sql
SELECT *
FROM employees
WHERE department = 'IT'
ORDER BY hire_date DESC;
```

Q6. Create and Populate Sales Table

```sql
CREATE TABLE sales (
    sale_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    amount INT,
    sale_date DATE
);

INSERT INTO sales (sale_id, customer_name, amount, sale_date) VALUES
(1, 'Aditi', 1500, '2024-08-01'),
(2, 'Rohan', 2200, '2024-08-03'),
(3, 'Aditi', 3500, '2024-09-05'),
(4, 'Meena', 2700, '2024-09-15'),
(5, 'Rohan', 4500, '2024-09-25');
```

Q7. Display All Sales Records Sorted by Amount (Highest to Lowest)

```sql
SELECT *
FROM sales
ORDER BY amount DESC;
```

Q8. Show All Sales Made by Customer 'Aditi'

```sql
SELECT *
FROM sales
WHERE customer_name = 'Aditi';
```

Q9. What is the Difference Between a Primary Key and a Foreign Key?

Primary Key: Uniquely identifies each record in a table. Cannot be NULL and only one primary key per table.

Foreign Key: Links one table to another. Refers to a primary key in

another table and maintains referential integrity.

Q10. What Are Constraints in SQL and Why Are They Used?

Constraints are rules applied to table columns to ensure data accuracy and integrity.

Common Types: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, DEFAULT.

They are used to maintain data accuracy, prevent invalid data entry, enforce business rules, and maintain relationships between tables.